

**An investigation of the sex-specific genetic architecture of fitness
in *Drosophila melanogaster***

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Abstract

In dioecious populations, the sexes employ divergent reproductive strategies to maximize fitness and, as a result, genetic variants can affect fitness differently in males and females. Moreover, recent studies have highlighted an important role of the mating environment in shaping the strength and direction of sex-specific selection. Here, we measure the sex-specific outbred fitness of 357 lines from the *Drosophila* Synthetic Population Resource (DSPR) in two different mating environments. We analyze the data using three different approaches to gain insight into the sex-specific genetic architecture for fitness: classical quantitative genetics, genomic associations, and a mutational burden approach. The quantitative genetics analysis finds that, on average segregating genetic variation in this population has concordant fitness effects both across the sexes and across mating environments. We do not find specific genomic regions with strong associations with either sexually antagonistic (SA) or sexually concordant (SC) fitness effects, yet there is modest evidence of an excess of genomic regions with weak associations, both with SA and SC fitness effects. Our examination of mutational burden indicates stronger selection against indels and loss-of-function variants in females than males.

Introduction

The genetic architecture of fitness variation (e.g., the number and identity of loci, and their effect sizes in different contexts) remains poorly understood. Given that any locus in the genome has the potential to contribute to fitness, the genetic architecture of fitness is likely highly polygenic. However, this does not preclude the possibility that some variants have reasonably large effects. Investigating the genetic architecture of fitness may provide useful clues as to the forces responsible for the maintenance of segregating fitness variants (e.g., mutation-selection balance or various forms of balancing selection; Mackay 2001; Barton and Keightley 2002; Frankham 2009)

In dioecious populations, the genetic architecture of phenotypic traits, including fitness may differ substantially between the sexes. Due to anisogamy, the sexes maximize fitness in different ways: in many systems, male fitness is limited by the number of mates that they acquire whereas female fitness is limited by the amount of resources that they can acquire and efficiently convert into offspring (Bateman 1948; Parker 1979; Wade 1979; Wade and Arnold 1980). Thus, traits necessary to maximize fitness, and by extension the genetic architecture underpinning those traits, can differ across the sexes.

Sex differences in the genetic architecture of fitness depend on the sex-specific effect size and direction of segregating variants. For example, some variants may have sex-limited effects because they are sex-linked or contribute to the expression of sex-limited traits (e.g., seminal fluid proteins). Such variants can be important contributors to the variance in fitness in one sex but may have little or no influence on fitness in the other. Variants with purely sex-limited fitness effects may be relatively rare. Instead, alleles may often contribute to variance in fitness in both sexes. The evolutionary fate of such variants will depend on the strength and direction of selection they experience when expressed in each sex. In some cases, alleles at a locus may have opposing,

sexually antagonistic, effects on fitness resulting in so-called ‘intra-locus sexual conflict’ (reviewed in Bonduriansky and Chenoweth 2009). Under specific conditions, such variants can be maintained by balancing selection, particularly when the effect sizes of variants are large (Kidwell et al. 1977; Connallon and Clark 2012). However, most genes are thought to serve similar functions in both sexes and disruptions to those functions are likely deleterious for both sexes (Rowe and Houle 1996; Whitlock and Agrawal 2009). While sexually concordant selection should remove genetic variation at these sites (Long et al. 2012; Connallon and Clark 2013), this class of loci may serve as a large mutational target and thus contribute substantially to the standing genetic variation in fitness via mutation-selection balance.

The primary objective of this study is to better understand fitness variation within and across the sexes. Ideally, one would catalog every variant and measure its fitness effect in each sex, but that ideal is not logistically feasible. Instead, we pursue three different approaches that each offer a different lens through which to view the relationship of the genetic architecture for fitness across the sexes.

First, we used a classic quantitative genetics approach to provide a summary of how genetic architecture of fitness is related between the sexes. This approach has been used in a number of laboratory (e.g., Vieira et al. 2000; Wayne et al. 2001; Pischedda and Chippindale 2006; Long et al. 2009; Collet et al. 2016) and wild (e.g., Kruuk et al. 2000; Merilä and Sheldon 2000; McCleery et al. 2004; Foerster et al. 2007) populations and is useful for asking how much additive genetic variation in sex-specific fitness exists in a population. In addition, we used this approach to estimate the intersexual genetic correlation for fitness (r_{mf}), which serves as a useful indicator of the overall concordance or antagonism of segregating genetic variation in a population. This correlation can be used to quantify the extent to which improvement in the fitness of one sex will

help or hinder the improvement of the other, at least in the short term. Empirical estimates of r_{mf} have been mixed: although a survey of animal species found that, on average, estimates of r_{mf} were small but positive (Poissant et al. 2010), several studies have reported negative estimates of r_{mf} of substantial magnitude (e.g., Chippindale et al. 2001; Fedorka and Mousseau 2004; Brommer et al. 2007; Punzalan et al. 2014). The power and limitation of the intersexual correlation is that it is a summary statistic that is predictive of short-term evolution that neither requires nor reveals information about individual allelic effects. Indeed, the underlying ‘genetic details’ matter little with respect to short-term evolution but can substantially affect evolutionary outcomes on longer time-scales.

Next, we used genomic association mapping techniques in an attempt to gain resolution on specific loci underlying sex-specific fitness. Association mapping provides a route to identifying individual allelic effects and, in principle, can be used to find the number and magnitude of mutations with sexually concordant, antagonistic, and sex-limited effects. Recognizing that the genetic basis of fitness is likely to be highly polygenic, we are unlikely to have the statistical power to identify a multitude of small effect variants. However, we were particularly interested in identifying loci contributing to sexual antagonism and such variants maintained by balancing selection are expected to be of larger effects (Kidwell et al. 1977; Connallon and Clark 2012), possibly detectable via genome association mapping. This approach has recently proven fruitful in identifying candidate sexually antagonistic loci in this species (Ruzicka et al. 2019).

Though association mapping is severely limited in its ability to identify small effect variants, other approaches can provide insights. When sequence and fitness data are both available for a collection of genotypes, then one approach is to quantify the aggregate effect of specified classes that are suspected to affect fitness *a priori* (Yang et al. 2017; Brown and Kelly 2019). For

example, one can use this approach to ask how the total number of loss-of-function variants (LoFs) in a genome affects fitness, which is equivalent to quantifying the average strength of selection against segregating LoFs. We are particularly interested in comparing the selection strength in males versus females. Strong competition for access to females has been postulated to increase the strength of selection on males relative to females on sexually concordant variants (reviewed Whitlock and Agrawal 2009), potentially increasing population mean fitness by reducing mutation load (Manning 1984; Agrawal 2001; Siller 2001) and accelerating adaptation (Lorch et al. 2003). Experimental studies using phenotypic marker mutations (e.g., Whitlock and Bourguet 2000; Pischedda and Chippindale 2005; Stewart et al. 2005; Sharp and Agrawal 2008; Hollis et al. 2009) and mutation accumulation lines (e.g., Enders and Nunney 2010; Mallet et al. 2011; Mcguigan et al. 2011; Sharp and Agrawal 2013; Grieshop et al. 2016) tend to support the notion of stronger selection in males than females, though there are potential concerns with these approaches (see Discussion).

The goals of this study were twofold. The first was to more fully understand the sex-specific genetic architecture of fitness using the three approaches described above. The second goal of this study was to investigate how the genetic architecture of fitness of each sex, as well as the relationship of these architectures across the sexes, was affected by the ecological context in which sexual interactions occur. Here, we estimated the sex-specific outbred fitness for 357 *Drosophila melanogaster* recombinant inbred line (RIL) haplotypes from the *Drosophila* Synthetic Population Resource (DSRP; King et al. 2012) in two mating environments: traditional fly culturing vials and larger cages that represent lower density and more structurally complex environments. Previous work has shown males and females interact differently in environments similar to the two used here (Yun et al. 2017; MacPherson et al. 2018). Male harm and sexual

conflict are strong features of sexual interactions in standard fly vials but not in the more complex cages; these differences affect the rates of nonsexual adaptation and purging (Singh et al. 2017; Yun et al. 2018).

Our quantitative genetic analysis recovers appreciable levels of additive genetic variation in fitness for both male and female fitness and finds that the fitness consequences of this genetic variation tends to be concordant across the sexes as well as across environments. Our association mapping analysis did not reveal any statistically convincing associations between fitness and any specific genomic region. However, there is some evidence that both sexually concordant and sexually antagonistic regions exist. Our mutation burden analysis recovers a significant negative association between the burden of putatively deleterious variants and female, but not male, fitness in both mating environments.

Methods

DSPR Inbred Genotypes

Recombinant inbred line (RIL) genotypes used for measuring sex-specific outbred fitness were obtained from the *Drosophila* Synthetic Population Resource (DSPR). The creation of these lines has been described in detail elsewhere (King et al. 2012a). Briefly, eight parental genotypes from geographically disparate regions were crossed in a round robin design to generate an outbred population that was allowed to recombine for 50 generations. Inbred genotypes were generated from this outbred, synthetic population following 25 generations of fullsib mating. Following low-resolution RADseq genotyping of inbred genotypes, a hidden Markov model (HMM) was implemented to determine the probable parental ancestry at regularly spaced 10kb intervals (or

haplotype blocks) along the genome of each line. The parental ancestry of the vast majority of 10kb blocks in most lines can be assigned to a parental source with high certainty. Here, we estimated sex-specific fitness of 357 lines. Although the majority of lines were assayed for fitness in both sexes and both environments, this was not true of all lines for logistical reasons (see Table S1 for number of lines assayed per sex and environment).

Measuring Fitness of DSPR Genotypes

To measure sex-specific outbred fitness of focal genotypes we first crossed females of each focal genotype to males from a common outbred population that was generated by mixing all DSPR founder genotypes. Focal males and females were then assayed in separate competitive fitness assays. Specifically, for each outbred genotype, we combined four focal males/females with twelve competitors of the same sex from an outbred laboratory population homozygous for a dominant *bw^D* marker. Focal and competitor genotypes competed for access to sixteen unmarked individuals of the opposite sex that came from an outbred laboratory population. For logistical reasons the source population for these individuals differed between male and female fitness assays: in male fitness assays we used females from an outbred laboratory population originally captured from a West African population (*Dahomey* stock) while in the female fitness assays we used males from an outbred stock generated by mixing DSPR founder genotypes.

All flies were collected at approximately the same age (~2-3 days post eclosion) and held separately by sex in vials sprinkled with live yeast in groups of sixteen for 24 hours prior to the start of fitness assays. All flies for a replicate were then combined and allowed to interact for 6 days in either a standard *Drosophila* culture vials (28 x 95 mm, approximately 60 mL) that contained 7ml of food sprinkled with live yeast or a larger, complex mating cages. The cages were polypropylene ULINE® deli containers (114.3 x 76.2 mm, approximately 500 mL) with two food

cups containing 7mL of food each, sprinkled with live yeast. Replicates were maintained in a 12L:12D light cycle at 25°C. Fresh media, sprinkled with live yeast was provided on day one and three (fresh media was not provided on day three in cage assays due to logistical reasons). On day six, all females were removed and, to reduce subsequent larval densities, divided approximately evenly into two vials sprinkled with live yeast and allowed to oviposit for ~24 hours. Starting ten days later, and repeating for a total of three days, offspring from the two oviposition vials were collected, pooled and frozen daily. The vast majority of offspring emerged on the first two days of collection. Following collection, we randomly sampled at least 100 offspring and calculated the proportion of total offspring with wildtype (red) eye colour in this sample. We have five replicate assays for most combinations of line, sex, and environment. Due to human error, some lines have more than five replicates while others have fewer. Fitness assays were conducted in a blocked design; for analyses, fitness estimates were standardized by dividing by the block mean fitness for that sex and environment and zero centered.

Quantitative Genetic Analysis of Fitness in the DSPR

To estimate the sex-specific heritability in fitness in each mating environment, we fit separate mixed models for each mating environment to estimate the sex-specific additive genetic and residual variances. We fit the model $W_{ijk} = \mu + \phi_{jk} + e_{ijk}$ where W_{ijk} is the within-block standardized and zero-centered fitness of the i th replicate of sex j of genotype k (i.e., RIL k). μ is the grand mean relative fitness of all lines, ϕ_{jk} is the random genetic effect of sex j of genotype k and follows a bivariate Gaussian distribution such that $\phi_{jk} \sim N(0, \mathbf{G})$ where

$$\mathbf{G} = \begin{bmatrix} \sigma_{RIL:m}^2 & \sigma_{RIL:m,f} \\ \sigma_{RIL:m,f} & \sigma_{RIL:f}^2 \end{bmatrix}$$

is a genetic covariance matrix. The diagonal elements are the variances among RILs for male and female fitness, which are proportional to the additive genetic variances; the off-diagonal is the among-RIL intersexual covariance, which is proportional to the additive genetic covariance. (Our estimated matrix is proportional, rather than equal, to the traditional $\mathbf{G}$ matrix because our unit of observation is the fitness of groups of four focal individuals sharing a RIL haploid genome rather than a single individual; see Supplemental Material for derivation). e_{ijk} specifies the residual variance of the i th replicate of sex j of genotype k and are identical and independently Gaussian distributed such that $e_{ijk} \sim N(0, \sigma_{ej}^2)$ where σ_{ej}^2 is the residual variance in fitness in sex j . As shown in the Supplementary Material, heritability of fitness in sex j as

$$h_j^2 = \frac{\sigma_{RIL:j}^2}{2 \left(\sigma_{ej}^2 + \frac{\sigma_{RIL:j}^2}{4} \right)}$$

The mixed models were fit using *MCMCglmm* (Hadfield 2010) separately for each mating environment. Models were fit using uninformative priors and run for 1,000,000 iterations after a burn-in period of 10,000 iterations with a thinning interval of 1000. Posterior estimates of variance components from each run were used to calculate the mean and 95% credibility intervals (CI) for each parameter of interest (i.e., $\sigma_{RIL,m}^2$, $\sigma_{RIL,f}^2$, h_m^2 and h_f^2). Specifically, CIs reported were calculated using the *HPDInterval* function in *MCMCglmm*. Estimates of variance components were also used to estimate the intersexual genetic correlation for fitness (r_{mf}) separately within each mating environment as

$$r_{mf} = \frac{\sigma_{RIL:m,f}}{\sqrt{\sigma_{RIL:m}^2 \sigma_{RIL:f}^2}}$$

Though our estimates of r_{mf} are intended to represent the intersexual genetic correlation with respect to “adult” fitness, they are indirectly affected by juvenile survivorship because we assess the fitness of adult males and adult females by measuring the number of offspring that survive to the adult stage. Genetic variation for offspring survivorship will upwardly bias our estimate of the intersexual correlation for adult fitness. However, we expect this effect to be small for several reasons. First, we attempted to keep offspring density from being high so that survivorship should be high. Second, the contribution of the genetic variance in offspring survivorship to the variance in focal adult fitness will be diluted by 50% because offspring only have half the genes from the focal parent. Third, previous studies have indicated that genetic variation for juvenile survivorship is small, especially compared to that of adult fitness components (e.g., Leips and Mackay 2000; Sharp and Agrawal 2018).

As a separate question, we were interested in the relationship between the genetic architectures of a given sex across the two mating environments. Using the fitness data for a single sex from each of the two environments, we employed a similar approach to estimate the within-sex cross-environment genetic correlation for fitness (i.e., $r_{env,m}$ and $r_{env,f}$ for males and females, respectively).

Association Mapping of Fitness and Indices of Sexual Antagonism and Concordance

We briefly outline methods for genome association mapping and provide details below. To identify genome associations with sex-specific fitness as well as loci underpinning sexually antagonistic and concordant variation, we first identified univariate traits for fitness and indices of sexual antagonism and concordance. Next, we defined a statistical model for mapping genotypic

variation to trait variation and a significance threshold was established to identify loci putatively associated with fitness or axes of antagonism/concordance.

To map phenotypic variation in male and female fitness we calculated the mean relative fitness, separately for each sex, of each RIL. To associate loci with sexual antagonism and sexual concordance, we defined an ‘axis of antagonism’ (i.e., SA Axis) and an ‘axis of concordance’ (i.e., SC Axis) following the method by Berger et al. (2014). Specifically, the matrix of coordinates of mean relative male and female fitness of each RIL are rotated such that estimates are projected onto axes that are 45° and 315° about the horizontal, which correspond to the axis of concordance and antagonism respectively. The value of a RIL along the axis of antagonism is calculated as:

$$SA = x \cos(45^\circ) - y \sin(45^\circ)$$

where x and y are the estimates of mean female and male relative fitness. RILs with above (below) average values for female fitness but the opposite for male fitness are expected to have positive (negative) values of SA. The value of a RIL along the axis of concordance is computed as:

$$SC = x \sin(45^\circ) + y \cos(45^\circ)$$

RILs with above (below) average fitness for both sexes are expected to have positive (negative) values of SC. Our primary interest was with respect to the SA index, but we independently mapped four traits—SA index and SC index, as well as male and female fitnesses—separately for each environment (i.e., 8 total traits).

A feature of the DSPR is that the mosaic haplotype structure of each RIL is known to a high degree of certainty in most cases. Specifically, at 10kb intervals along the genome of each RIL, the additive probability of each of the eight parental genotypes is known from a Hidden Markov Model (King et al. 2012a). To associate loci with each of the eight traits we regressed, at

each 10kb haplotype block, the RIL trait values against the additive probability for each of the eight parental lines. Specifically, we fit the mixed model:

$$\bar{Z}_i = \mu + \sum_{j=1}^l \beta_j G_{ij} + \psi_i + e_{ij}$$

where $\bar{Z}_i$ is the mean trait value of the i th RIL and μ is the grand trait mean across all sampled RILs, G_{ij} is fixed effect of the additive probability of the haplotype block of the i th RIL having descended from parental genotype j , β_j is the additive genetic effect of the j th parental genotype on the trait, ψ_i is the random effect of RIL genotype that is assumed to follow a Gaussian distribution such that $\psi_i \sim \sigma_{kl} \sim N(0, \mathbf{K})$ where $\mathbf{K}$ is the relatedness matrix of the DSPR lines (see below), l is the number of parental genotypes (max = 8), that contributed a cumulative additive probability of at least 5% at each haplotype block i across all RIL genotypes (see below) and e_{ij} specifies the residual variance of the i th RIL at the j th haplotype block and is assumed to follow a Gaussian distribution with a mean of zero and unknown variance. At each 10kb haplotype block i , we excluded any RIL genotypes whose most probable parental ancestor did not have a cumulative additive probability of at least 5% among all RIL genotypes. P -values were obtained from a likelihood ratio test against a null model where $\beta_j = 0$ for all j .

To estimate the genetic relatedness matrix of lines assayed, we calculated the average pairwise relatedness for all 357 inbred genotypes over all haplotype blocks. Specifically, for each pair of genotypes i and j we calculated the genetic relatedness (κ_{ij}) as

$$\kappa_{ij} = \frac{\sum_{n=1}^m \sum_{k=1}^8 p_{ikn} p_{jkn}}{m}$$

where p_{ikn} is the additive probability that haplotype block (i.e., a 10kb interval) n in genotype i originated from ancestral genotype k and m is the total number of haplotype blocks where genotypic information is present in both inbred genotypes.

As reported below, we were unable to find strong statistical support for individual haplotype blocks contributing to sexual concordance or antagonism. A plausible explanation is that effect sizes are low relative to environmental variation, so it becomes unlikely to find the very low p -values required when controlling for multiple testing. To explore the data, we shifted our focus to testing for an excess of haplotype blocks with p -values below a given critical value, p_{crit} . In principle, the expected number of such blocks under the null hypothesis of no true genetic effects (i.e., the number of false positives, x) should be $E[x] = p_{crit}n_{total}$ where n_{total} is the number of blocks tested. If the tests were independent, it would be straightforward to calculate the full probability distribution of x under the null hypothesis and properly test whether the observed number of ‘significant’ blocks $n_{p \leq p_{crit}}$ is significantly larger than expected under the null hypothesis. There are two potential issues with this approach. First, it relies on the p -value from each individual test accurately reflecting the chance of a false positive under the null hypothesis (i.e., that all the assumptions of the mixed model are well met). Second, the tests are not independent due to linkage disequilibrium. Permutations offer a conceptually simple (though computationally intensive) solution to both issues. We permuted line identity (i.e., genome-wide genotype) with respect to the set of fitness values (i.e., the 8 traits) and reanalyzed the data. From each permutation, we recorded the p -value for each block (and for each trait) and then calculated an x value (i.e., number of ‘significant’ blocks) for a given statistical criteria ($p_{crit} \in \{0.001, 0.01, 0.05, 0.1\}$). To reduce computational time, we only tested every 7th block; the same blocks were tested in the unpermuted and permuted data sets. (There is little additional information

gained from testing all blocks because of the high linkage disequilibrium among neighboring blocks.) 2077 permutations were performed.

False discovery rates are an alternative approach to the multiple testing problem inherent in genomic analyses but are not necessarily a perfect solution. The reliability of these methods depend not only on the validity of the underlying p -values, but can depend on the independence of tests as well as the shape of the distribution of p -values. The R package *qvalue* (Storey et al. 2021) was used to convert p -values into q -values (i.e., FDR rate) based on a previously described method (Storey and Tibshirani 2003). We calculated q -values with the unpermuted data as well as each permuted data set. As we did with p -values, we compare the number of blocks with a q -value below a given threshold, q_{crit} , to the distribution under the null hypothesis as determined from the permutations.

Gene Ontology (GO) Enrichment Analysis

We conducted a gene ontology (GO) enrichment analysis of genes found in “genomic clusters” that are associated with male and female fitness and the SA and SC axes. To help control for strong linkage between loci, we employed a clustering algorithm to define genomic clusters that are associated with each trait. For each chromosome arm, our clustering procedure began at the block furthest away from the centromere. We used a sliding window of 3.5 Mb to identify windows (or clusters) that had at least one haplotype block with $p < 0.1$. When a significant haplotype block was identified, we examined the immediately proceeding 3.5 Mb window. If this window contained any haplotype blocks that reached statistical significance, these blocks were subsumed into the previous cluster. This procedure was repeated until no haplotype blocks in the

proceeding 3.5 Mb window reached statistical significance. Thus, a cluster was defined as a set of statistically significant haplotype blocks that are closer than 3.5 Mb to each other. After obtaining genomic clusters associated with each of the eight traits (see above), we used Ensembl's *Variant Effect Predictor* (VEP; McLaren et al. 2010) to find genes that fall 5Kb upstream or downstream of the leading haplotype block in each cluster (i.e., the haplotype block with the lowest p -value). This analysis used a liberal $p < 0.1$ cut off for statistical significance of haplotype blocks but in many cases the leading block for a cluster would have a much lower p -value. Next, we used *GOrilla* (Eden et al. 2009) to test for an enrichment of GO terms in genes associated with each trait relative to a background set of all genes annotated in the *Drosophila* genome from our VEP analysis. p -values for GO terms were converted to q -values and a q -value threshold of 0.05 was used to assess statistical significance.

Selection on Putatively Deleterious Variants in the DSPR lines

Our pipeline to identify putatively deleterious variants (i.e., single nucleotide polymorphisms [SNPs] and insertion/deletion [indel] variants) involved three steps. First, we imputed the underlying genomic sequence at sites segregating for SNPs and indels in the DSPR for all 357 focal genotypes assayed in this study. Following quality control steps, we attempted to enrich for sites likely to be deleterious by filtering out any variants segregating in either of two geographically disparate outbred populations. Finally, we mapped the variants to functional categories and retained only those variants falling within coding regions and whose functional consequence could reasonably be expected to be deleterious.

Genomic Imputation of Inbred Genotypes

To impute SNPs segregating among inbred genotypes, we used the results of a HMM implemented to uncover the parental ancestry of each genotype at regularly spaced 10kb intervals along the genome of each inbred genotype and a table of SNPs segregating in each parental line from the DSPR webpage (<http://wfitch.bio.uci.edu/~dspr/>). We filtered out all sites in the parental SNP table where less than 95% of SNP calls were to either the reference or alternate allele in each of the parental lines, sites with less than 15 reads and all multiallelic sites. Within each line, we filtered out any 10kb intervals where the ancestry assignment probability of the most probable of the eight parental lines was less than 95%. This filtering step removed an average of ~8.5% of blocks from each RIL.

Imputation of insertion/deletion (indel) variants began by obtaining indel VCFs for each of the eight DSPR parental lines from the *Drosophila* Genome Nexus (DGN) webpage (<https://www.johnpool.net/genomes.html>). We filtered out indels not supported by at least 30 reads. We then imputed indels in the 357 DSPR lines in the same manner as described above for SNPs.

Identifying putatively rare variants from the DSPR

Because deleterious variants are expected to segregate at low frequency at mutation-selection balance, a reasonable approach to identifying putatively deleterious variants might include focusing only on low frequency variants. To this end, we filtered out any variants segregating in the DSPR that were also segregating in two geographically disparate outbred populations of *D. melanogaster*: 197 genotypes from Zambia (*Drosophila* Population Genomic Project Phase 3 [DPGP3]) and 205 genotypes from Raleigh, North Carolina (*Drosophila* Genetic Reference Panel [DGRP]).

FASTA files containing whole genome sequences for all individuals sampled in both the DPGP3 and DGRP populations were obtained from the DGN webpage. These genome sequences were generated following a standardized alignment and quality control procedures described elsewhere (Lack et al. 2015, 2016). Using Perl scripts supplied on the DGN webpage, we further filtered sites that showed evidence of high relatedness and recent admixture. We concatenated all genomes in a single population into a single multi-FASTA alignment file and generated a VCF containing SNPs for each population using *SNP-sites* (Page et al. 2016). Next, we filtered out any SNPs in the DSPR that were present in VCFs for either of the two outbred populations (i.e., sites segregating for variants in the DSPR that were also captured as segregating in the two outbred population samples were excluded from downstream analyses).

Next, we needed to determine which variant in the DSPR was the major allele in nature. We chose one genome at random from the DPGP3 (ZI-103) and one genome at random from the DGRP (RAL-100). For both genomes, we filtered out any sites that were polymorphic within the populations that they were sampled from and then filtered out any sites that differed between these two sequences including all sites that were masked in either sequence (these sites were also filtered out from the DSPR). We compared this sequence (which we term the ‘outbred consensus sequence’) back to the DSPR VCF file described above and filtered out any sites where one of the two segregating alleles in the DSPR was not identical to the outbred consensus sequence. Finally, for each site in the DSPR, we deemed the variant matching the outbred consensus sequence as the major allele.

Indels were filtered from DSPR lines in a similar manner. We obtained indel VCFs for both outbred populations from the DGN webpage and combined each individual VCF using the ‘*vcf-merge*’ command in *VCFtools* (Danecek et al. 2011). Finally, we filtered out any indels in the

DSPR lines that shared a starting coordinate with indels segregating in the either the DGRP or DPGP3 indel VCFs.

Functional annotation of genomic variants in the DSPR and identifying putatively rare deleterious variants

We predicted the functional consequences of minor allele SNPs and indels in the DSPR lines using VEP (McLaren et al. 2010). Genomic coordinates for all VCFs were updated to release 6 of the *Drosophila* reference genome using the UCSC liftOver tool (Kent et al. 2002) prior to passing VCFs to VEP. We retained only variants that fell within coding regions. For SNPs, we focused on two classes of variants: those variants labelled by VEP as high ‘IMPACT’ and missense mutations resulting in radical amino acid changes. Variants with high IMPACT scores include variants predicted to have severe functional consequences including splice acceptor/donor disrupting variants and stop loss/gain and start loss variants. Following Sohail et al. (2017), we refer to this combined set of high IMPACT variants as ‘loss-of-function’ (LoF) mutations. For each missense variant we characterized the resulting amino acid change as either a conservative or radical change based on an amino acid classification scheme proposed by Zhang (2000) that classifies amino acids into six categories that vary with respect to polarity and volume. An amino acid change is deemed ‘radical’ if the major and minor alleles code for amino acids that are in different categories with respect to either polarity or volume. Radical substitutions are under stronger purifying selection in *Drosophila* than conservative substitutions (Smith 2003). Finally, we use VEP to map indels to the reference genome and filtered out all indels that did not map to coding regions.

In summary, we obtained a list of three classes of putatively rare variants segregating in coding regions among the sample of DSPR lines assayed in our study: LoF mutations, variants

resulting in radical amino acid changes, and indels. This list of variants served as putatively deleterious variants segregating in the DSPR. Finally, we scored each inbred genotype for the number of variants of each class that they carry. We termed the number of putatively deleterious variants of carried by a genotype to be that genotype’s “mutational burden”. Note that the estimated number of deleterious variants for each RIL will, on average, be slightly below the true number because a small fraction of genomic blocks are excluded from each RIL. The “uncounted” variants are a small source of measurement error in the estimate of burden but are not expected to create biases in the effects of interest (i.e., direction and magnitude of variants or sex differences in these effects). Two genotypes were clear outliers with respect to the number of SNPs causing radical amino acid changes (Figure S1) and as result these two genotypes were removed from further analysis.

Statistical Analyses

The mutational burden for genotype j is represented by the numbers of LoFs (X_j), radical amino acid variants (Y_j), and coding region indels (Z_j). We modelled fitness as function of these three components of mutational burden. Our interest was in whether we could detect fitness effects of these mutational burdens and whether these effects may depend on sex or environment. Because of the large number of possible interaction terms, we began with a model selection approach to compare four models encompassing our major interests. In the “full” model (Model 1), we fit the following mixed model:

$$W_{ijk} \sim S_i + V_k + X_j + Y_j + Z_j + S_iX_j + S_iY_j + S_iZ_j + V_kX_j + V_kY_j + V_kZ_j + S_iV_kX_j + S_iV_kY_j + S_iV_kZ_j + \phi_j + e_{ijk}$$

Here, W_{ijk} is the mean relative fitness of the i th sex of the j th genotype assayed in mating environment k . Relative fitness was standardized to the within-block mean and zero centered; for each line, the average fitness across replicates was used for this analysis. S and V indicate sex and mating environment, respectively. The model includes genotype as a random effect (ϕ_j) that is assumed to follow a Gaussian distribution with a mean of zero and unknown variance. e_{ijk} specifies the residual error of the i th sex of the j th genotype assayed in mating environment k . In Model 2, we excluded the three three-way interactions (i.e., sex-by-environment-by-burden component) but retained all six two-way interactions. In Model 3, we retained the interactions between sex and each burden component but excluded all interactions involving the environment. In Model 4, we retained the interactions between environment and each burden component but excluded all interactions involving the sex.

Statistical analyses were conducted in *R* (version 3.6.2; R Development Core Team 2019). Models were fit using the *lmer* function of the *lme4* package (Bates et al. 2015) after loading *lmerTest* (Kuznetsova et al. 20) so that t -values were calculated using Satterwaite's method. We confirmed the significance of the reported results by comparing each parameter estimate to a null distribution created from analyses in which we permuted the set of fitness values (for each sex and environment) with respect to genotype (RIL ID and three burden components).

Results

Quantitative Genetic Analysis

We recover modest additive genetic variation (σ_{RIL}^2) for fitness in both sexes when assayed in simple (Figure 1; $\sigma_{RIL,f}^2 = 0.020$ [95% C.I.: 0.015, 0.025]; $\sigma_{RIL,m}^2 = 0.027$ [0.020, 0.035]) and complex ($\sigma_{RIL,f}^2 = 0.032$ [0.023, 0.040]; $\sigma_{RIL,m}^2 = 0.035$ [0.025, 0.046]) mating environments. We

confirmed, by visual inspection of quantile-quantile plots of posteriors distributions, that the posterior distributions for all variance components estimates were approximately Gaussian and not right skewed. Estimates for the heritability h^2 in fitness are modest in males (simple environments: $h_m^2 = 0.102$ [0.074, 0.134]; complex environments: $h_m^2 = 0.124$ [0.089, 0.161]) and in females (simple environments: $h_f^2 = 0.150$ [0.111, 0.186]; complex environments: $h_f^2 = 0.164$ [0.121, 0.208]).

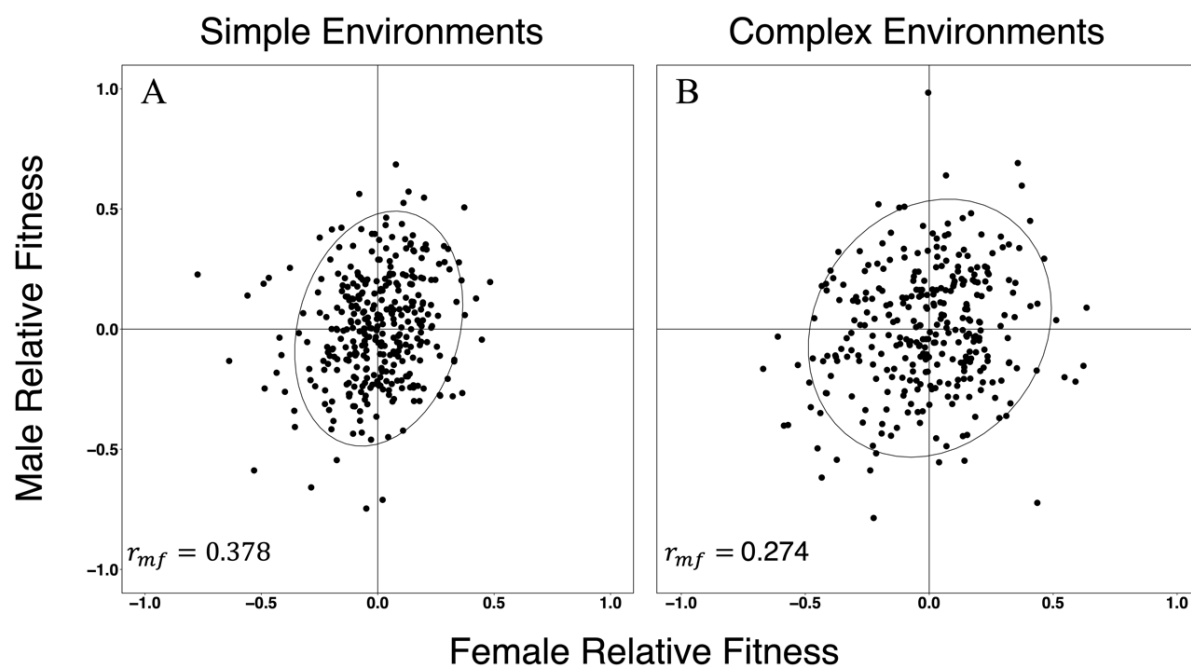


Figure 1: Male and female mean relative fitness of DSPR genotypes measured in (A) simple and (B) complex mating environments. Fitness measures are standardized to the within-block mean fitness for each sex and environment respectively and zero centered. Ellipses represent the 95% confidence ellipse for the distribution of sex-specific mean fitness estimates.

The intersexual genetic correlation for fitness was positive and significantly different from zero in both mating environments (Figure 1; simple: $r_{mf} = 0.378$ [0.158, 0.583]; complex: $r_{mf} = 0.274$ [0.086, 0.465]). Though significantly positive, these correlations are far from unity, suggesting a fair degree of genetic independence between the sexes. The cross-environment

correlation for fitness was also positive and significantly different from zero in both sexes (Figure 2; $r_{env,m} = 0.423$ [0.246, 0.636]; $r_{env,f} = 0.322$ [0.150, 0.471]).

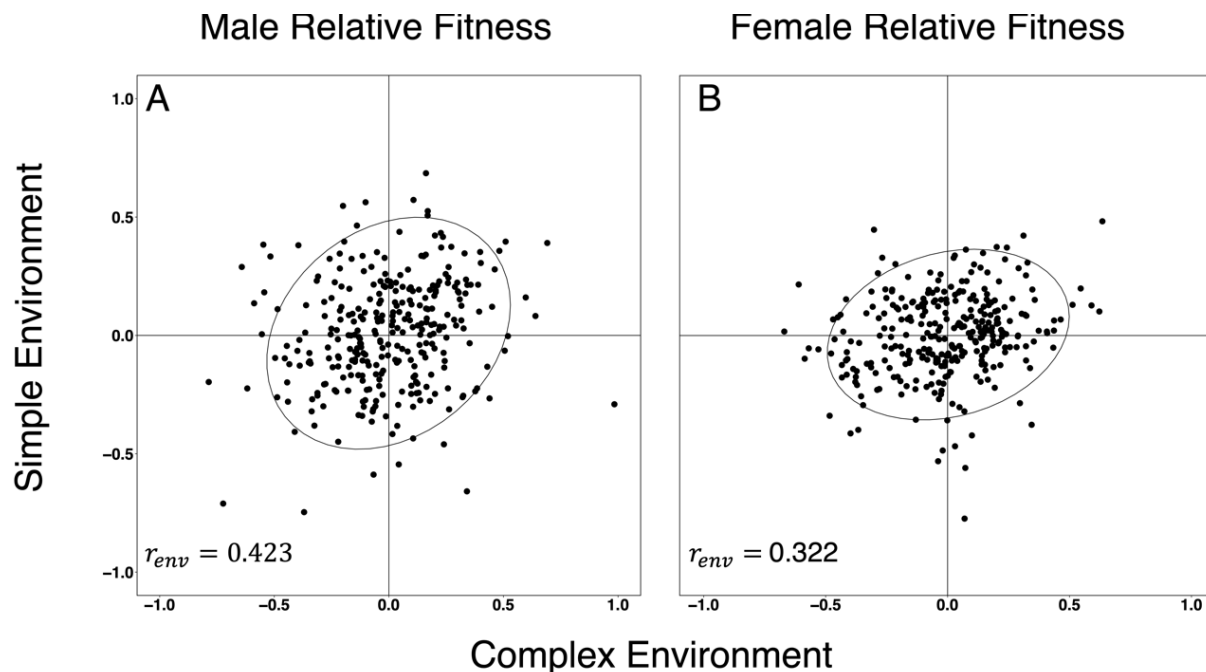


Figure 2: Mean relative fitness measured in simple and complex mating environments in (A) males and (B) Females. Fitness measures are standardized to the within-block mean fitness for each sex and environment respectively and zero centered. Ellipses represent the 95% confidence ellipse estimated from the sex-specific mean fitness estimates.

Association Mapping of Sexually Antagonistic and Sexually Concordant Loci

We present the results for sexual antagonism (SA) and sexual concordance (SC) here and results for male and female fitness are given in the Supplemental Materials. Manhattan plots for SA and SC in the simple and complex environments are shown in Figure 3. Few haplotype blocks have strong statistical associations with SA or SC; thus, we have little confidence in specific regions affecting SA or SC. Consequently, we shifted our focus to exploring whether there is any evidence at all of regions associated with SA or SC.

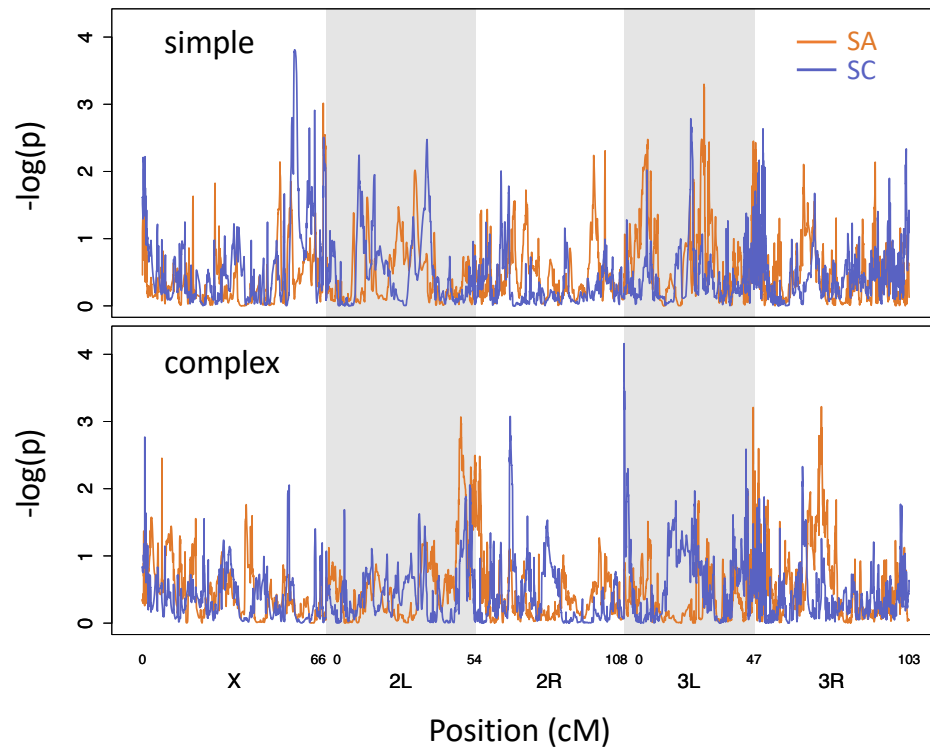


Figure 3: Manhattan plots. Strength of statistical support for associations between 10kb haplotype blocks along the genome with sexual antagonism (SA; orange) and sexual concordance (SC; purple) in DSPR lines when fitness was measured in the simple (top) or complex (bottom) mating environment.

Given that the classical quantitative genetic analysis indicates the existence genetic variation, it is plausible that many blocks have modest effects that do not yield extreme p -values (e.g., $p < 10^{-5}$). To explore this possibility, we counted the number haplotype blocks $n_{p \leq p_{crit}}$ that passed a given statistical requirement (e.g., $p_{crit} = 0.001$, $p_{crit} = 0.05$) and compared this number to the distribution under the null hypothesis of no genetic effects as generated by permutation.

Consider the results using $p_{crit} = 0.05$. We first ask if there is evidence of genetic associations of any sort, considering the tests for all four traits (i.e., SA and SC in both environments). We find 610 tests that pass the criterion of $p \leq 0.05$ (“significant blocks”). In permutations, the average number of significant blocks is about half this number (353.3) and fewer than 1% of permutations result in 610 or more blocks (i.e., there are more associations than expected by chance). Examining SA and SC separately, there are 338 significant associations with SA considering both environments. In permuted data sets, the average number of SA blocks is 176.2 and fewer than 2% of permutations have 338 or more. There are 272 significant SC blocks across environments though ~10% of permutations have as many or more. Examining SA and SC in each of the environments separately, there are more observed blocks than the corresponding average number observed for each of the four traits, but only for SA in the simple environment is the observed number in the upper 5% of the tail of its corresponding permutation distribution (Table 1).

These results are sensitive to the statistical criterion (p_{crit}) used to identify significant blocks (Table 1). In most cases, the observed number of blocks is greater than expected (i.e., the means of the permutation distributions) but whether the excess is substantially greater than expected depends on which trait is being considered and what statistical criterion is used. Whereas the results based on $p_{crit} = 0.05$ indicate an excess of associations for SA but not SC, with $p_{crit} = 0.001$

or $p_{crit} = 0.1$, there appears to be an excess of associations for SC but not SA. With $p_{crit} = 0.01$, neither type of association is in the upper 5% of the tail of the permutation distribution.

The use of permutations as a basis for a null distribution is important in evaluating the data from the DSPR. As an example, consider the associations tests for SA in the simple environment using $p_{crit} = 0.05$. 1682 blocks are tested. Under the null hypothesis of no true associations, we expect $1682 * 0.05 = 93.1$ ‘significant’ blocks on average; the mean of the permutations is close to this value at 88.1. If the tests were independent, then the variance among permutations should be $1682 * 0.05 * 0.95 = 88.5$; however, the observed variance—1952—is much larger. This higher variance is likely due to the non-independence of tests because of linkage disequilibrium, which can result in the false positive rate being much higher (or much lower) than expected under independence in any given instance (i.e., individual permutation or real data). The use of q -values does not solve this issue (Table S9). For example, applying the significance criterion of $q \leq 0.3$, only 73% of permutations result in 0 ‘significant’ blocks and, of the remaining 27% of permutations, the average number of ‘significant’ blocks is 129 (range: 1-1675).

553 Table 1. The number haplotype blocks passing a specified statistical criteria. For this analysis, only every 7th haplotype block was tested (1682
554 blocks tested). n is the observed number in the real (i.e., unpermuted data). $\bar{n}_{perm}$ is the mean number of associations in permuted data sets.
555 P_{perm} is the proportion of permutations with as many as or more than the observed number of associations (n); cases where the observed value
556 is the upper 5% of the distribution (i.e., $P_{perm} < 0.05$) are shown in bold. Results using a q -value criterion are shown in the Supplementary
557 Material, as are results for male and female fitness (Tables S2-4).

Criterion for 'significant' association	Metric	Any trait	SA	SC	SA in simple environment	SC in simple environment	SA in complex environment	SC in complex environment
	n	24	3	21	3	7	0	14
$p_{crit} = 0.001$	$\bar{n}_{perm}$	7.5	3.9	3.7	2.1	1.8	1.7	1.9
	P_{perm}	0.056	0.396	0.025	0.232	0.079	1	0.025
	n	117	54	63	34	34	20	29
$p_{crit} = 0.01$	$\bar{n}_{perm}$	73.2	36.2	36.9	18.3	18.7	18.0	18.3
	P_{perm}	0.117	0.184	0.131	0.148	0.152	0.333	0.201
	n	710	338	272	195	162	143	110
$p_{crit} = 0.05$	$\bar{n}_{perm}$	353	176	178	88.1	89.1	88.0	88.5
	P_{perm}	0.0087	0.018	0.095	0.031	0.084	0.127	0.274
	n	1064	479	585	261	298	218	287
$p_{crit} = 0.1$	$\bar{n}_{perm}$	699	348	351	173	174	174	176
	P_{perm}	0.0077	0.091	0.016	0.100	0.055	0.230	0.062

Following the procedure outlined in the Methods to generate lists of candidate regions, we conducted GO enrichment analyses (Tables S5-S12). Under the statistical criterion used, no GO terms were enriched among genes associated with the SA and SC in the simple environment. However, we find evidence of enrichment of GO terms among genes associated with SA and with SC in the complex environment. For SA, the enriched categories are biological processes that affect basic cell function as well as higher levels of regulation (e.g., post-translational regulation, epigenetic regulation; Table S7). For SC, a single molecular process was enriched: dipeptidyl-peptidase activity (Table S8). Though the terms enriched are not readily interpretable, the existence of enriched terms is another line of evidence that the association mapping results are more than mere noise.

Mutational Burden

The assayed DSPR genotypes had on average 5.6 (range: 1-12) LOFs, 274.8 (39-363) radical amino acid missense variants, and 125.7 (75-172) indels in coding regions. We first compared four models that differed in their inclusion of interactions of burden with sex and/or environment. Based on AIC, the best of these models included interactions of burden with sex but burden not with environment (“Model 3”). The results of this model are shown in Table 2 (see Tables S13-S16 for results of the other three models and their AICs). Figure 4 shows the model fit relationship between mutation burden and fitness for the three classes of putatively deleterious variants from Model 3. This model indicates significant selection against LoFs and indels but also significant interactions of these two burden components with sex. In both cases, the interaction term indicates that selection is significantly stronger against the burden component in females than

males. Because of the significant interaction terms involving sex, we subsequently fit models for each sex separately. These models were specified in the same way but all terms with ‘sex’ were omitted. In females there is a significant negative relationship between relative fitness and both LoF burden and indel burden (Table S16). Female fitness is reduced by $\sim 1\%$ per LoF and $\sim 0.1\%$ per indel. There is no significant association of radical amino acid burden in females. In males, we cannot detect significant associations between male fitness and any component of mutational burden (Table S17).

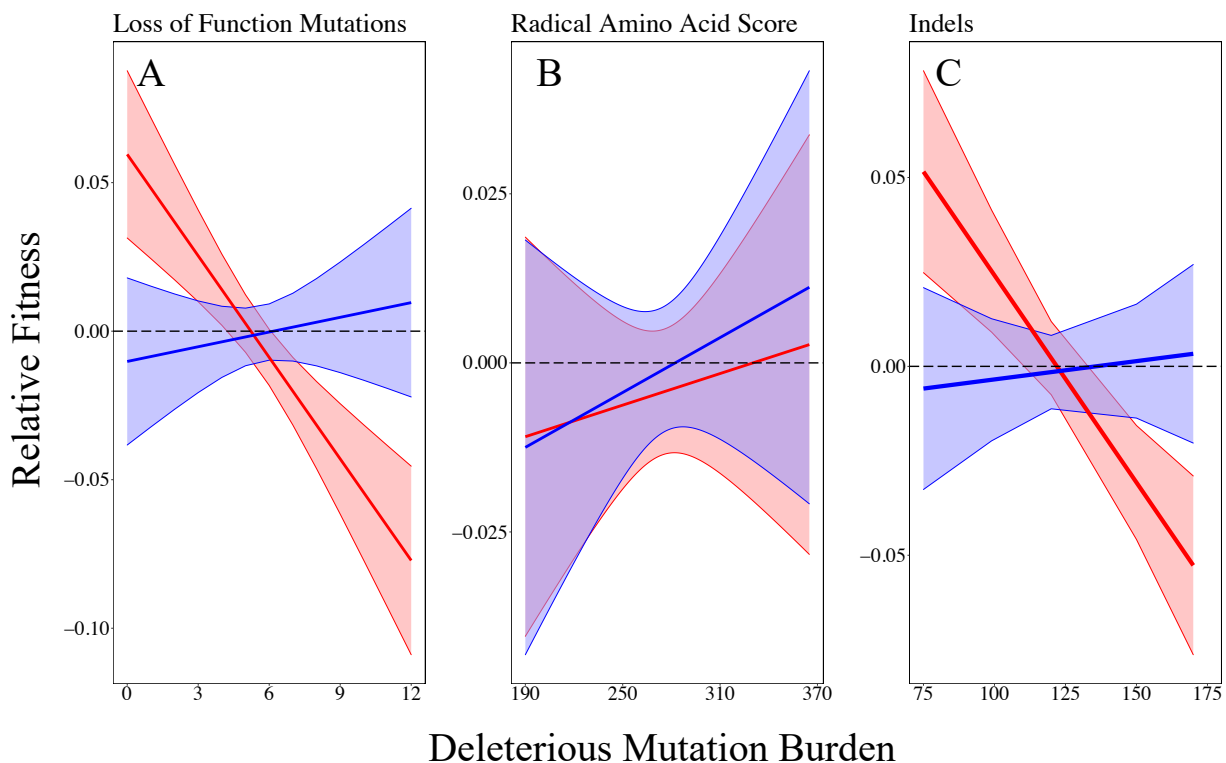


Figure 4: Model fits of the relationship between deleterious variant burden of (A) loss-of-function (LoF) mutations (B) radical amino acids (C) coding region indels and male (blue) or female (red) relative fitness (after zero-centering relative fitness). Shading represents the 95% confidence intervals for model fits.

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597 **Table 2:** Parameter estimates from mixed model fit to evaluate the fitness effects of each of three
 598 components of mutational burden—loss-of-function (LoF) burden, radical amino acid burden, and
 599 indel burden—and their interactions with sex. Bold terms indicate statistically significant effects
 600 at $\alpha = 0.05$.

Effect	Estimate	Standard Error	<i>P</i>
Intercept	0.17	0.11	0.10
Sex (Male)	-0.24	0.13	0.08
Environment (Simple)	0.005	0.011	0.69
LoF Burden	-0.011	0.005	0.02
Radical Amino Acid Burden	0.00008	0.00033	0.81
Indel Burden	-0.0011	0.0005	0.03
LoF Burden × Sex (Male)	0.013	0.006	0.03
Radical Amino Acid Burden × Sex (Male)	0.00006	0.00041	0.89
Indel Burden × Sex (Male)	0.0012	0.0006	0.05

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Discussion

The sexes represent different selective environments: surveys of phenotypic selection have found that shared traits experience persistent sex differences in both the strength (Singh and Punzalan 2018) and direction (Cox and Calsbeek 2009; De Lisle et al. 2018) of selection. Thus, there is considerable scope for segregating genetic to affect fitness of the two sexes differently. On the other hand, the two sexes share many similar requirements so selection on many variants will be similar. In this study, we used three approaches to explore different aspects of the sex-specific genetic architecture of fitness in a ‘synthetic’ laboratory population of *Drosophila melanogaster*.

Quantitative Genetics Analysis

Our quantitative genetics analysis recovers significant levels of additive genetic variance for fitness in both sexes in the two mating environments where we assayed fitness. The heritabilities of fitness were modest in both sexes in the two mating environments. These results are similar in magnitude as reported in other laboratory populations of *Drosophila* (e.g., Fowler et al. 1997; Gardner et al. 2005; Pischedda and Chippindale 2006; Long et al. 2009; Ruzicka et al. 2019) and other insect species (e.g., Archer et al. 2013; Berger et al. 2014) but our estimates of the heritability of fitness are much higher than those reported from studies of wild populations (e.g., Gustafsson 1986; Kruuk et al. 2000; Merilä and Sheldon 2000; McCleery et al. 2004; Teplitsky et al. 2009; McFarlane et al. 2014). Low heritability estimates in wild populations are likely the result of high levels σ_E^2 in fitness (Price and Schluter 1991; Houle 1992) rather than a lack of genetic variation. In addition, the population used in our study has only experienced ~50 generations of adaptation to laboratory conditions in which we measured fitness. Thus, there may be more fitness-

affecting variants segregating at intermediate frequency relative to an adapted population where directional selection is expected to have reduced σ_A^2 .

The intersexual genetic correlation for fitness r_{mf} gives us a genome-wide summary of the extent to which the shared genome constrains the sexes from reaching their sex-specific trait optima (Lande 1980) at least in the short term. Our estimates of r_{mf} were positive and significantly different from 0 in both mating environments (Figure 1). This result implies that, on average, genetic variants have sexually concordant fitness effects and, as a result, evolution is not strongly constrained from improving the fitness of both sexes simultaneously, at least in the short-term. Though estimates for r_{mf} in both environments are significantly greater than 0, they are also far from 1. This intermediate value for r_{mf} suggests that while sexually concordant variants may be predominant, other types of variants (e.g., sex-limited and/or sexually antagonistic variants) are likely also segregating. This view is consistent with our genomic association mapping analysis (though the latter should be regarded with caution because of limited power): the Manhattan plots (Figure 3) suggest that sexually antagonistic genomic regions are present and are only modestly less frequent than sexually concordant ones.

r_{mf} is not an immutable property of a species or a population. The evolutionary history of a population in relation to ecological context in which fitness is measured can substantially alter r_{mf} . For example, the genetic variation in populations where both sexes are approaching their sex-specific fitness optima may become enriched for genetic variants with sexually antagonistic fitness effects as selection will have eroded sexually concordant genetic variation (Long et al. 2012; Connallon and Clark 2013), resulting in a negative r_{mf} . However, in a novel environment where the fitness landscape is perturbed, the phenotypic mean values for both sexes may be above or

below the new fitness optima and thus segregating genetic variation may experience sexually concordant selection resulting in a more positive r_{mf} . The ‘synthetic population’ used here was created from mixing isogenic lines and allowing for 50 generations of recombination and, presumably, lab adaptation (King et al. 2012a,b). Thus, it is plausible that our estimates of positive r_{mf} reflect an overall concordance of selection on average as a consequence of this population being relatively maladapted to the laboratory conditions in which we measured fitness, which was not identical to how the lab population was originally created by King et al. (2012a,b).

Our significantly positive estimates of r_{mf} stands in contrast to a classic study using the same species by Chippindale et al. (2001) who reported a significantly negative r_{mf} . That study measured fitness in a lab context to which their fly population was well adapted. However, not all studies neatly fit within the narrative in which the degree of adaptedness explains variation in r_{mf} . Sharp and Agrawal (2018) estimated a positive though non-significant values for r_{mf} using another *D. melanogaster* population that was well-adapted to the lab; assay conditions were similar to but did not exactly replicate maintenance conditions. A particularly interesting case is that of Collet et al. (2016) who measured r_{mf} in two descendant populations of the fly population studied by Chippindale et al. (2001). Despite being maintained and assayed similarly, the two descendant populations had significantly different values of r_{mf} , with r_{mf} significantly negative in one and non-significantly positive in the other. That study demonstrates that r_{mf} can evolve substantially over a few hundred generations in ways that are not completely captured by “adaptedness”.

In addition to evolutionary changes in r_{mf} (i.e., changes in the underlying genetic architecture), r_{mf} can be directly affected by the environment. Studies have found that r_{mf} may be highly sensitive to the ecological context in which genetic variation is expressed (e.g., Berger

et al. 2014; Punzalan et al. 2014; Duffy et al. 2019). However, our estimates of r_{mf} in the two mating environments are reasonably similar (and confidence intervals are broadly overlapping). Given that sexual interactions are quite different in these two types of environments (Yun et al. 2017), it was not obvious from the outset that the two r_{mf} estimates would be so similar.

The similarity of r_{mf} estimates would be unsurprising if the genetic architecture of fitness for each sex was the same across the two environments (i.e., if r_{env} was close to 1 for both sexes). Indeed, r_{env} was significantly positive for both sexes. Nonetheless, both correlations were substantially less than 1, indicating that the genetic architecture for fitness differs somewhat across environments. This is not surprising given that these two environments likely select for different phenotypic traits in both sexes. We have particularly strong reasons to believe this for males. In a different experiment, populations that evolved for a few tens of generations in environments similar to those examined here had diverged significantly for male fitness in a genotype-x-environment dependent manner (Yun et al. 2019), presumably because these alternative mating environments modulate the opportunity for male harassment to result in successful mating (Yun et al. 2017). In contrast to males, females showed much weaker signs of divergence following adaptation to these alternative environments. As such, it is perhaps surprising that here we have found the cross-environment genetic correlation is of similar magnitude for both sexes.

It is notable that the intersexual genetic correlations within an environment (r_{mf}) and the cross-environment genetic correlations for each sex (r_{env}) are all significantly positive and of similar magnitude. As mentioned above, a simple explanation for all correlations being positive is that this population contains an excess of variants that are unconditionally deleterious in the lab environment, perhaps because of its history (i.e., insufficient time to adapt to general lab

conditions). Although particular suites of phenotypes may maximize fitness for each sex and mating environments, it is plausible that many of the same morphological, physiological and behavioural traits necessary to acquire fitness are similar between the sexes in both environments. Nonetheless, it is surprising that r_{mf} and r_{env} estimates are of similar magnitude; roughly speaking, this suggests that the differences between sexes are as large as the differences between environments with respect to segregating variation. For example, knowing that a genotype confers high fitness to males in one environment is approximately as good a predictor of conferring high fitness to females in that same environment as it is of conferring high fitness to males in the other environment.

Association Mapping Analysis

Our association mapping analysis revealed few, if any, genomic regions with very strong statistical associations with the traits examined. Consequently, we took a more exploratory approach to evaluating whether there was any evidence of genetic associations beyond what would be expected by chance, using permutations to create appropriate null distributions. The results of these analyses were somewhat inconclusive. On the one hand, in almost all cases the observed number of associations was larger than expected. However, the permutation-based null distributions have high variance and, in only some cases, was the observed number of associations unusually high relative to the corresponding null distribution. With some criteria, there was a substantial excess of associations with SA but not SC; with other criteria, there was an excess of associations with SC but not SA.

It is clear our power is limited. In general, the power to detect a significant genomic association is inversely related to the minor allele frequency (MAF) at loci underlying trait

variation. An advantage of this synthetic population is that the variants segregating in this panel should have high MAFs (i.e., a neutral expectation of 12.5%) compared to natural populations. However, in reality, the observed MAFs are often lower than 12.5%, presumably due to 50 generations of drift and selection. Of course, statistical power is also directly related to effect size. Because of selection, the standing variation will be depauperate of variants with large directional effects (e.g., unconditionally deleterious) segregating at high frequency. It is likely there are numerous variants that make small contributions to fitness variation, but we would not expect to detect them via this association mapping approach. Nonetheless, it seemed plausible we could detect large-effect sexually antagonistic variants if such alleles were segregating. Theory indicates that large-effect sexually antagonistic variants are more readily maintained by balancing selection than small effect ones (Kidwell et al. 1977; Connallon and Clark 2012). Our failure to observe any strong associations implies that the loci underlying fitness are unlikely to be of large effect.

Our tentative interpretation of the association study is that effect sizes are small to moderate for both SA and SC as we tend to see more modest associations than expected yet fail to find very strong associations. In addition, these data suggest that SA variation is not particularly rare relative to (detectable) SC variation as we see only modestly stronger evidence for associations with SC than SA (Table 2). Broadly speaking, these interpretations are consistent with results from the classical quantitative genetic analyses that indicate genetic variance for both male and female fitness and only a modest positive genetic correlation.

Using a permissive threshold to generate larger lists of candidate regions, we found some evidence of enriched GO terms for SA and SC in the complex but not the simple environment. In their genomic analysis of sexual antagonism, Ruzicka et al. (2019) found little evidence of enriched GO categories and those they did where not obviously linked to sexually dimorphic traits.

One interpretation of these results is that sexual antagonism is diffuse affecting many types of processes and/or is especially prone to persisting in genes affecting a variety of basic functions that are under selective constraint. However, another interpretation is that studies such as ours lack power to meaningfully identify genomic regions experiencing sexually antagonistic selection. On the other hand, even though the terms identified are not readily biologically interpretable, the existence of significantly enriched GO terms for both SA and SC in the complex environment provide another form of evidence that the association analysis is not just noise.

Mutational Burden

We find evidence of selection against mutation burden in females but not males. All of the statistically significant estimates of selection are, reassuringly, negative (i.e., selection disfavours variants that we *a priori* expected to be deleterious). Examining entire classes of variants, rather than individual variants, allows us to detect smaller effects. Specifically, we find a negative association between female relative fitness and both LoF burden and indel burden, with each additional LoF reducing female fitness by, on average, ~1%, and each additional indel by ~0.1%. In contrast, we find no significant associations between any of the three classes of putatively deleterious variants and relative fitness in males in either mating environment. These results imply that selection acting on mutational burden is stronger in females than in males.

These results are surprising considering many studies have found evidence for stronger selection acting on variants expressed in males. Several studies have estimated the sex-specific selection acting on marker mutations in *Drosophila* (Whitlock and Bourguet 2000; Pischedda and Chippindale 2005; Stewart et al. 2005; Sharp and Agrawal 2008; Hollis et al. 2009). These studies

have generally found evidence in support of the hypothesis that the strength of selection is male biased but because visible marker mutations are unlikely to be representative of the typical mutations that arise in a population, it is difficult to generalize these results. Several studies examining the fitness consequences of novel mutations via mutation accumulation (MA) experiments in *Drosophila* and other species (e.g., Enders and Nunney 2010; Mallet et al. 2011; Mcguigan et al. 2011; Sharp and Agrawal 2013; Grieshop et al. 2016) have also generally supported the hypothesis of stronger selection through males. While an advantage of using MA lines is that they capture the fitness effects of more ‘typical’ mutations, one disadvantage is that the causative mutations are typically completely unknown. Fitness results from MA studies may be dominated by a small subset of the mutations that have large effects. This is problematic if such mutations are unrepresentative in other ways. For example, perhaps mutations with male-limited effects are disproportionately likely to be of large effect.

It remains unclear why our analyses suggest stronger selection in females than males whereas past studies using phenotypic marker mutations or MA lines found the opposite. Our study focuses on variants with specific molecular characteristics. Perhaps more importantly, the genetic variation in the DSPR has been filtered by selection in the wild in contrast to MA studies that capture the fitness effects of all novel mutations (excluding only extremely large effect variants e.g., lethal, or sterilizing variants). Variants causing strong effects in the wild (in either sex) are expected to be underrepresented relative to mutational input. One possibility for the discrepancy of experimental results would be that fitness effects on males are reasonably consistent between field and lab conditions whereas effects on females are more variable between environments. Thus, variants with large fitness effects in males in the lab would be underrepresented whereas those causing large female effects in the lab would be not be so underrepresented due to past selection

in the wild. Perhaps selection is more consistent on males between field and lab because their fitness is so heavily determined by their interactions with other flies via mate competition whereas female fitness may be more directly affected by the environment itself (e.g., temperature, humidity, food availability).

A second possibility for the discrepancy between this and past study is from differences in how fitness was assayed. First, viability is a potentially important source of fitness variation in our study but minimized in many other studies. We measured fitness of adults via offspring produced after a six day ‘mating interaction’ period whereas other fly studies typically use much shorter mating periods. Our protracted period of mating interaction allows for more fitness variation via adult mortality, especially so for females; males that die during this period can still contribute offspring via matings prior to death but females dying prior to the day we allowed for egg laying would prevent any contribution to offspring production in our study (i.e., fitness effect of mortality during the mating period is stronger on females than males).

While the type of variants we studied are often presumed to be deleterious, very few studies have attempted to validate this presumption by examining associations with fitness. Two recent studies have fruitfully employed similar methods to investigate the aggregate fitness consequences of putatively deleterious variants. Yang et al (2017) employed a comparatively conservative criteria to filter for putatively deleterious variants in maize (*Zea mays*) by filtering SNPs based on both their genomic evolutionary rate profiling (GERP) scores—a measure of evolutionary constraint on a SNP (Cooper et al. 2005)—and the predicted functional consequence of variants. Brown and Kelly (2019) took a more liberal approach to identify deleterious variants in yellow monkeyflower (*Mimulus guttatus*) by filtering out any variant with a minor allele cut-off greater

than 5%. Both studies we able to identify significant associations between mutation burden and measures of fitness.

Conclusion

Any study of the genetic variation in fitness should be considered with respect to the history of the population in relation to the environment in which fitness was assayed. Our study likely represents a modestly, but not exquisitely, well adapted population. Relative to an adapted population at equilibrium in a constant environment, we would expect the genetic lines used here to harbour more sexually concordant variation. Despite this expected “bias” towards concordance, both the quantitative genetic and association mapping results suggest sex differences in the genetic architecture of fitness. The observed intersexual genetic correlations are significantly greater than zero, yet they are also substantially less than 1. Within the limits of our statistical power, there is evidence for genomic regions that are sexually concordant but also regions that are sexually antagonistic. While the history of the population is expected to affect the relative amounts of concordant versus antagonistic variation, there is no *a priori* reason it should differentially affect the sexes with respect to the fitness effects of putatively deleterious variants. Nonetheless, we find evidence of stronger selection in females than males, contrary to studies of selection against phenotypic markers and new mutations. While caution should be used in interpreting these results, they add to the existing empirical evidence of sex differences in selective effects.

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